Pre-build 2. Solar-powered boat

Your goal is to design and construct a solar-powered boat that can travel the fastest across the pool.

The boat has to be propelled by an electric motor on board of the boat. Electric energy from the solar panel accrued during the trial can be the only source of energy used by the boat.

Your boat must be constructed using only wood, cardboard, aluminum foil and cling wrap (or similar thin clinging foil), except for the propeller shaft, motor, gears or pulleys (if needed), solar panel, electronic parts (if needed), and connecting wires. No ready-made boat can be used. Glue may be used for joining parts but cannot serve as a structural element. The boat's dimensions must not exceed 25 cm in height or length, and 15 cm in width. No energy of any kind can be stored in the boat before the start of the trial. The boat must have an easy way for the judges to verify that no energy has been stored.

Judges will provide a standard light source for all teams, shown here: https://www.amazon.com/AUYKTE-Rechargeable-Flashlights-Flashlight-Emergencies/dp/B0CH86RSC6

The length of this flashlight is 18.5 cm and the diameter of the narrow part is 3.3 cm. The light source will be set to a continuous high-beam setting. The team may adjust the focus of the beam only before the beginning of the trial. The light will be placed on a table that is 73 cm high.

The team should provide a stand to hold the light source and direct the light in the desired direction. The stand should be maximum 25 cm in height, length, or width including the mounted light source. The stand and the light have to remain on the table during the trial. The stand may allow for some movement of the entire light source during the trial, but this motion may only be powered by gravitational energy and cannot move the light outside of the allowed stand dimensions. In that case, the movement has to start simultaneously with switching the light on. No one may touch the flashlight, its stand, or the boat after the start of the trial. The stand cannot be attached to the table but can extend from the edge of the table and below the edge as long as the restrictions on the size are obeyed and it is sitting stable on the table. Any materials can be used to build the stand.

The trial will begin when the flashlight is turned on and will last for three minutes. The boat should move as fast as possible across the pool and finish at the latest by the end of the three-minute trial.

Your boat may incidentally contact the walls or the lane separator of the tank, but you may not utilize these for propulsion. The boat cannot touch the bottom of the tank at any time. The depth of the pool will be at least 30 cm

Scoring

The score will be the length of the pool in m divided by the time it took the boat to get across in seconds. If there is a tie, the heavier boat wins. If the boat does not reach the

end of the pool the score will be distance traveled along the long side of the pool in m divided by 300 s.

Each team has only one trial, and the preparation and the trial cannot take longer than 5 minutes from the time the team's time allocation begins.

Figure 1: Test set-up

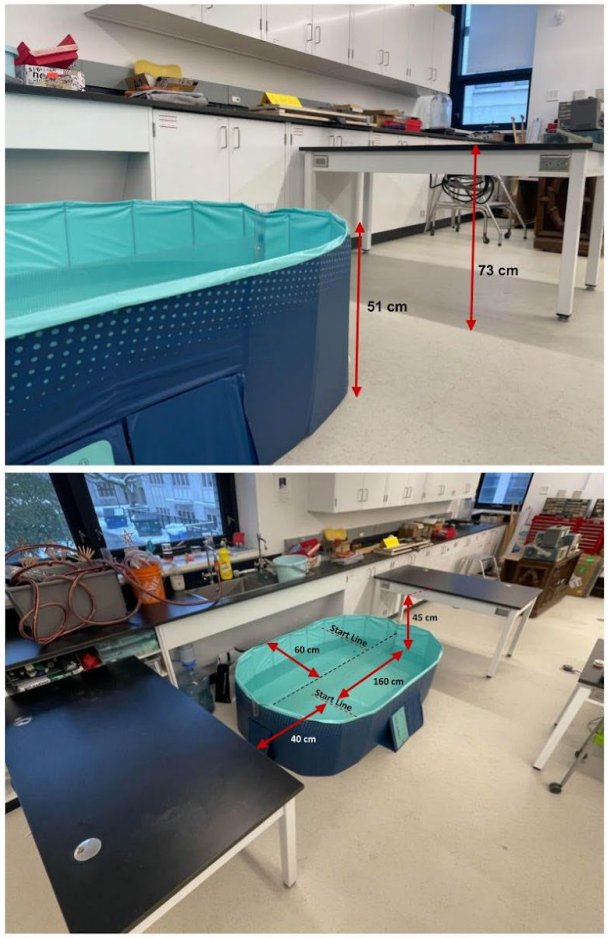

Figure 1: Horizontal distance from the edge of the table to the start line in the water is 40 cm. Vertical distance from the top table surface to the water surface is 45 cm. The distance from the floor to the top of the pool wall is 51 cm. The length of the pool inside is 189 cm. The pool is divided into two separate lines by a solid divider. There will be two boats tested simultaneously in opposite directions.